

Analysis of the 1MBO protein with homology modelling and molecular docking

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Abstract: This project focuses on the analysis of the Myoglobin (1MBO) protein using homology modelling and molecular docking to understand its interaction with the HEME ligand. The 1MBO structure was prepared by separating the protein chain from the ligand, calculating right after, the Solvent Accessible Surface Area (SASA) to analyse the protein-ligand interface in a deeper level. Homology models were generated with SWISS-MODEL, AlphaFold2 and ESM, which were then used to compare with the crystal structure of our protein to identify the best structural prediction. Then HADDOCK 2.4 was used for the molecular docking part of the work, where we evaluated different types of docking approaches, including: blind, random patches and active & passive residues to predict the binding modes. The last step consisted in identifying the Hotspots present in our complex, through Hotspot Wizard we were able to identify the key residues that influence protein stability and ligand interaction. All these findings provided valuable insights into the structural and functional properties of the Myoglobin.

Keywords: Homology Modelling; Hotspots; Molecular Docking; Myoglobin Protein; SASA

Introduction

In this work, our focus was on exploring core steps used in Proteomics for the myoglobin protein, which was the chosen macro-molecule for this project, using tools for protein preparation and analysis, for purposes of interface analysis, in our case for the HEME ligand, the Solvent Accessible Surface Area (SASA) calculations to better understand the characteristics of the binding between our protein-ligand complex, homology modelling and comparisons with different predicted models to evaluate structural prediction accuracy, molecular docking to see how the HEME ligand interacts with the chain A of the myoglobin ending with hotspot analysis.

The Myoglobin (1MBO) is a cytoplasmic hemoprotein that is present, only in vertebrate oxidative skeletal muscle fibers and cardiac myocytes, which have an important role on oxygen transportation (Gros et al., 2010; Ordway & Garry, 2004). Myocardial infarction is one of the associated coronary pathologies characterized by sudden cardiac death (Salari et al., 2023). Nowadays, Myoglobin is considered an older cardiac biomarker for the early detection of acute myocardial infarction and is now rarely used for that purpose due to the availability of newer and more specific markers (Dasgupta & Wahed, 2021).

The structure of myoglobin was originally delineated by Kendrew et al. in the year 1958, allowing later work to show that the backbone of myoglobin was a polypeptide chain that consists of eight α -helices (Figure 1A). It is also known that Oxygen (O) is bound to myoglobin through its HEME group, whose composition consists of iron ions and porphyrin rings. With the way the polypeptide chain is folded, the HEME group stays located between two Histidine residues, His64 and His93 (Figure 1B), while the iron ion coordinates have six ligands, of which, four of them are provided by the Nitrogen atoms of the four pyrroles, and the fourth ligand is provided by the Imidazole side chain of His93 which is known for coordinating the HEME group by pushing the iron ion slightly out of the plane of the HEME group. The sixth coordinate position, which is not occupied in Deoxy myoglobin is not only the oxygen binding site but for other ligands like CO or NO as well. Also, with the use of X-ray diffraction and myoglobin xenon structure binding have identified four internal cavities, and that positioning is also of molecules for interaction with the heme group (Frauenfelder et al., 2001; Ordway & Garry, 2004).

One of the biggest and most important software used throughout this study was PyMOL, which is a powerful tool for molecular visualization, usually used to analyse and prepare protein structures. It allows for the manipulation of molecular models, identification of key residues, and the preparation of input files for computational analyses, all procedures that were done for our 1MBO protein (Schrödinger & DeLano, 2020).

Another crucial part of this protein analysis was the understanding of Solvent Accessible Surface Area (SASA), which is crucial to understand the interactions and structural dynamics between different types of complexes, Antibody-Antigen, Protein-Protein, and in our case Protein-Ligand. SASA measures the surface area of a biomolecule that is accessible to a solvent like water, giving insights into residue exposure, hydrophobic interactions, and binding interfaces (Lee & Richards, 1971). For these calculations we used the PDBePISA, which is a web server that can be used to calculate solvent exposure values for monomeric and complexed forms of the proteins, which enables the determination of variation in SASA upon complex formation

(Δ SASA) and relative SASA normalized to the maximum possible value (RelSASA) (Krissinel & Henrick, 2007).

The Homology modelling consists in predicting the protein's 3D structure based on known and experimentally obtained structures of homologous proteins, these are proteins that share a common evolutionary background/origin, having similar amino acid sequences, structures and even functions. They have two types of classifications, they can either be orthologs, which originate from speciation and perform similar functions in different species; or be paralogs, resulting from gene duplications within the same genome and therefore may develop new functions. Homology, in summary, suggests a shared ancestry between these proteins. Having this definition in mind, this method then relies on evolutionary conservation of structures and is helpful for understanding molecular interactions, stability and functions (Schwede et al., 2003).

Molecular docking is a type of test used for predicting the preferred orientation of a ligand when bound to a target protein. Docking is a computational technique used to predict how two molecules, in our case a protein and a ligand, interact with each other. It involves simulating the binding process to identify the optimal orientation, position, and conformation of the ligand within the protein's active site, being widely used in drug discovery to assess molecular compatibility, affinity and stability of binding, under various conditions (Morris et al., 1998). HADDOCK 2.4 is a software that can be applied for this type of test, facilitating predictions of interactions through various docking approaches (Honorato et al., 2024).

Hotspots are regions on a protein's surface that play a crucial role in molecular interactions, such as binding with other proteins, ligands, or DNA. They consist of clusters of amino acids that contribute disproportionately to binding energy, making them, overall, a critical target for drug design. There are two types of Hotspots, they can be functional, where highlights residues critical for catalytic or binding activity, or stability, where reveal residues that contributes to the structural integrity of the protein (Pavelka et al., 2009). Hotspot Wizard identifies functional and stability hotspots within the interaction interface (Sumbalova et al., 2018).

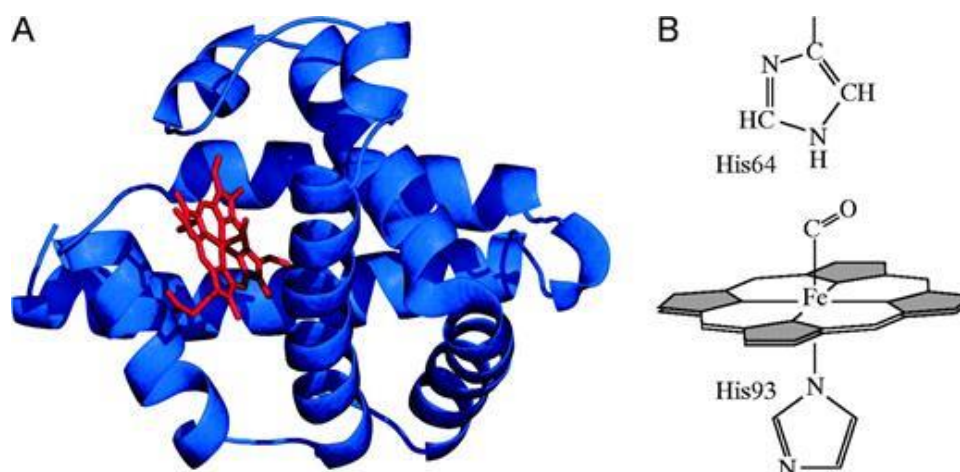


Figure 1 - A) Myoglobin 3D structure. The backbone of myoglobin consists of eight α -helices (blue) that wrap around a central cavity containing a HEME group (red), which is capable of binding various ligands including oxygen. **B)** Histidine residues above (His64) and below (His93). Adapted from Ordway & Garry (2004).

Analysis

Protein Analysis and Preparation

We downloaded the crystal structure for 1MBO from the Protein Data Bank (PDB). Using the PyMOL tool, our protein structure was prepared by removing the water molecules (solvent), adding hydrogens (H) to validate the structure (Figure 2), and separating our single protein chain, the HEME ligand and the full complex into individual PDB files (Figure 3). These prepared files, later served as inputs for subsequent analysis.

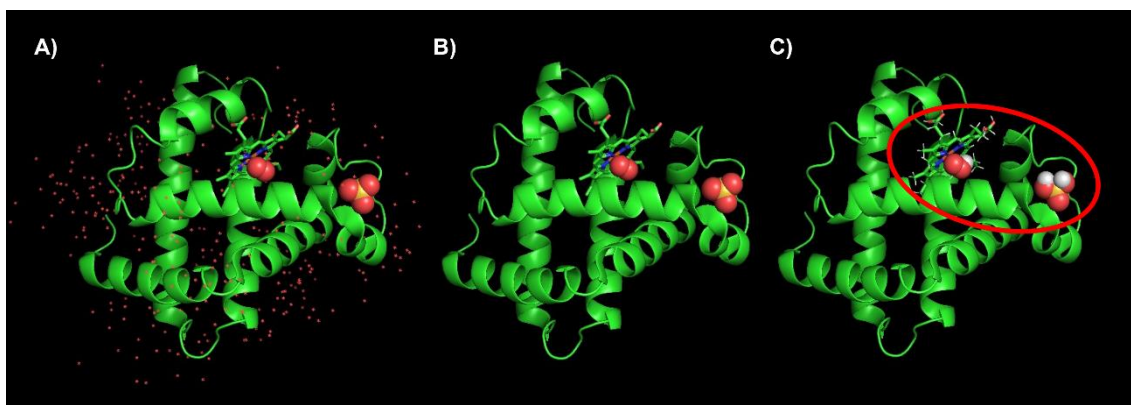


Figure 2 - 1MBO protein complex. **A)** original format of the molecule; **B)** removal of the solvent; **C)** HEME ligand with added Hydrogens

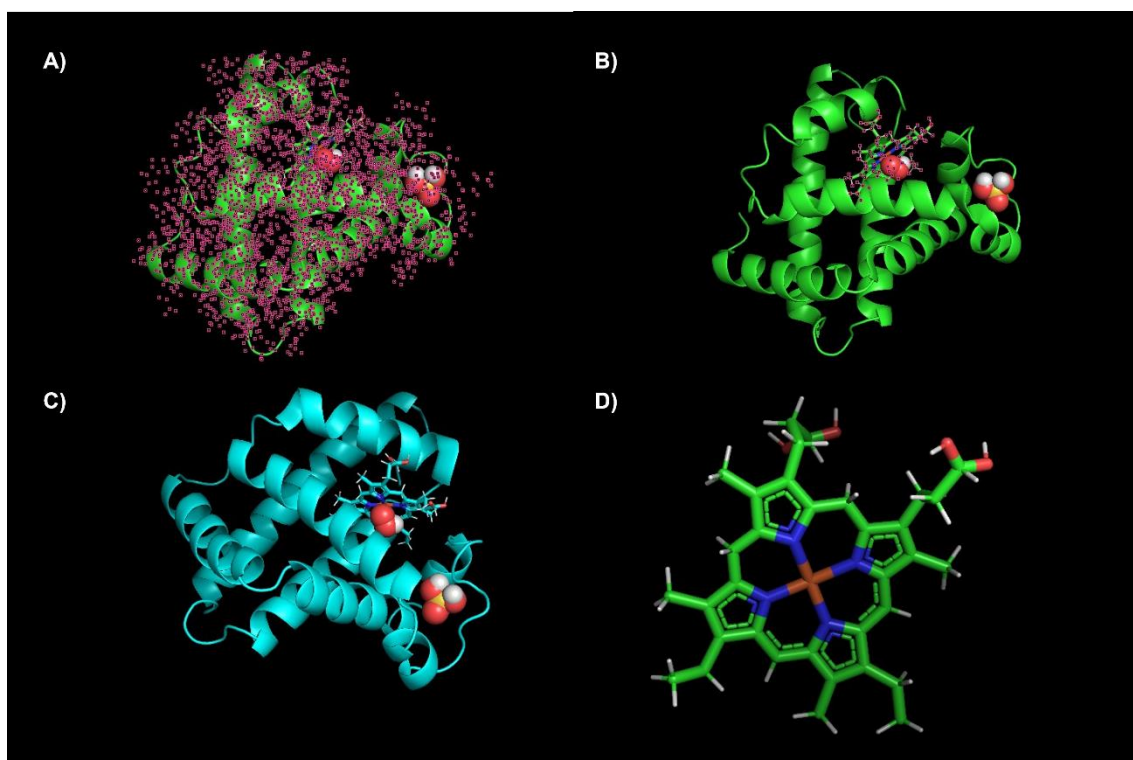


Figure 3 - 1MBO protein. **A)** selection of only the Chain A; **B)** selection of only the HEME ligand; **C)** protein-ligand complex molecule; **D)** HEME ligand.

Interface Analysis

The following step was really important for the next parts of the work, in which the residues involved in the protein-ligand interaction were identified. First, we started with PyMOL, which was used to visualise and manually identify the interface residues that were present within 5Å or less from the HEME ligand. Having identified a total of 30 different residues (R), their position (P), and the chain they were a part of (C). Then we were supposed to use the CoCoMaps web server to further confirm and validate the interfaces identified previously, however, the website is not working, so this step was unfortunately skipped.

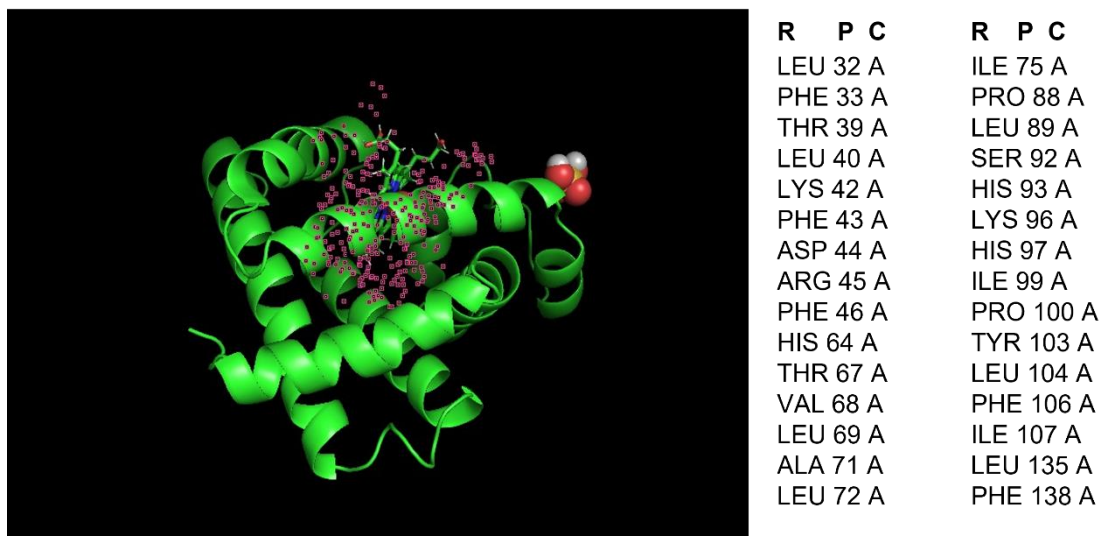


Figure 4 - 1MBO protein with the interface residues selected at 5Å distance from the HEME ligand with the list for them on the right.

Solvent Accessible Surface Area Calculations

Then, to measure the different Solvent Accessible Surface Area (SASA) values for the HEME ligand, the whole protein complex, and just the chain A, we used the PDBePisa web server. We chose this one as we were having conflicts with the original web server we chose, GetArea. We didn't even consider using PyMOL for these SASA calculations as we didn't know how to use it for that specific purpose at the time, so we opted for the next best option. For the identification of the CompSASA (complex) and MonSASA (monomer) values we submitted the PDB files for the HEME ligand, the whole complex and the chain A to PDBePisa, which returned us a list/table of values for each residue (Table 1). The calculation of the values for the Δ SASA (variation, delta) and RelSASA (relative) was done through the implementation of the formulas seen below in a python script created by us, which was able to extract the values from the table and execute the necessary steps automatically in order to obtain the variation and relative SASA (Figure 5).

Table 1 - Table with the name and number of residues and their respective CompSASA and MonSASA values obtained from PDBePisa.

Complex Summary

Multimeric state	1	Surface area, Å ²	7857.1	ΔG ^{int} , kcal/mol	-31.1	TΔS ^{diss} , kcal/mol	0.6	
Copies in unit cell	N/A	Buried area, Å ²	1448.8	ΔG ^{diss} , kcal/mol	12.5	Symmetry number	1	
Formula	Aabc						Biomolecule (R350)	1
Composition	A[SO4][HEM][OXY]							
Dissociation pattern	A[HEM][OXY] + [SO4]							

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##	Interfacing structures	N _{occ}	Diss.	Sym.ID	Buried area, Å ²	ΔG, kcal/mol	N _{HB}	N _{SB}	N _{DS}	CSS
1	⊙ [HEM]A:155 + A	1		1_555	551.0 (38%)	-19.9 (64%)	3 (38%)	0	0	0.061
7	○ [SO4]A:154 + A	1	x	1_555	77.9 (5%)	-11.3 (36%)	4 (50%)	0	0	0.038
10	○ [OXY]A:555 + A	1		1_555	54.9 (4%)	0.3 (1%)	1 (13%)	0	0	0.000
12	○ [OXY]A:555 + [HEM]A:155	1		1_555	40.6 (3%)	-0.2 (1%)	0 (0%)	0	0	0.000

View

Details

XML

Assembled monomers

##	Range	Surface area, Å ²	Buried area, Å ²	ΔG ^{int} , kcal/mol	N _{HB}	N _{SB}	Symmetry operation	Sym.ID	Transformation matrix			
									Rotation		Translation	
1	⊙ A	8182.0	683.8	-5.95	8	0	x,y,z	1_555	1.000	-0.000	0.000	0.000
									0.000	1.000	0.000	0.000
									0.000	0.000	1.000	0.000
2	○ [SO4]A:154	184.7	77.9	-11.23	4	0	x,y,z	1_555	1.000	-0.000	0.000	0.000
									0.000	1.000	0.000	0.000
									0.000	0.000	1.000	0.000
3	⊙ [HEM]A:155	818.9	591.6	-15.31	3	0	x,y,z	1_555	1.000	-0.000	0.000	0.000
									0.000	1.000	0.000	0.000
									0.000	0.000	1.000	0.000
4	○ [OXY]A:555	120.4	95.6	1.37	1	0	x,y,z	1_555	1.000	-0.000	0.000	0.000
									0.000	1.000	0.000	0.000
									0.000	0.000	1.000	0.000

A) $\Delta\text{SASA} = \text{monSASA} - \text{compSASA}$

$$\text{relSASA} = \frac{\Delta\text{SASA}}{\text{monSASA}}$$

B) This is the 1MBO Protein's deltaSASA value: 7498.2 and this is the RelSASA value: 0.9164263016377414
This is the 1MBO Heme's deltaSASA value: 267.9 and this is the RelSASA value: 0.3271461716937355
This is the 1MBO Complex's deltaSASA value: 6408.3 and this is the RelSASA value: 0.8156062669432742

Figure 5 - A) Formulas for ΔSASA and RelSASA; B) calculation of these values for both the HEME ligand and the protein-ligand complex for our 1MBO.

Homology Modelling

For the execution of the Homology Modelling, we started by obtaining the amino acid sequence of our 1MBO protein, which was extracted in FASTA format from the PDB and submitted first to the SWISS-MODEL web server to generate a homology-based 3D structural model. For this first method, the model was selected based on two metrics, the Global Model Quality Estimation (GMQE) and Quality Mean (QMEAN) scores. Then, to be able to make a conclusion on the best predictive structural tool, this same process was repeated for other platforms, more specifically, AlphaFold2 and ESM Metagenomic Atlas, in which both evaluate the confidence score and quality of its models on the pLDDT value, which is similar to the B factor, where the quality of the model is shown based on colour, the more confident it is the more blue the structure will be, and the less confident the more red it will be. Once we had these three predicted 3D structures, we went back to PyMOL to compare them with our crystal structure by individually aligning each predicted structure with the crystal one and calculating the RMSD values for each. Once all calculations are made, we then check for the alignment that showed the lowest RMSD value, as it is the indicator of the most similar structure, therefore, the best predicted structure.

First, we submitted our Fasta file for the 1MBO protein to all the three web servers for 3D structure prediction, SWISS-MODEL, AlphaFold2 and ESM. When it came to the SWISS-MODEL we had some initial issues, as the RMSD for the model we chose for alignment, had a value of 19.654, which definitely wasn't right, so we ran the test once again and saw that we made a mistake when choosing the best model, and after some examining we found the best one, which had the highest levels of confidence, a GMQE close to 1 and a QMEAN close to 0.7, which after aligning with the crystal structure ended up having a way better RMSD score of 0.396 (Figure 6). Following the SWISS, we had the ESM model that had a high confidence value pLDDT > 90, and the worst RMSD score in comparison, at around 0.412. The model predicted by AlphaFold2 also had a high score of confidence pLDDT > 90, exhibiting the lowest RMSD value at 0.381, therefore being the one with the best structural prediction for the subsequent docking results (Figure 7).

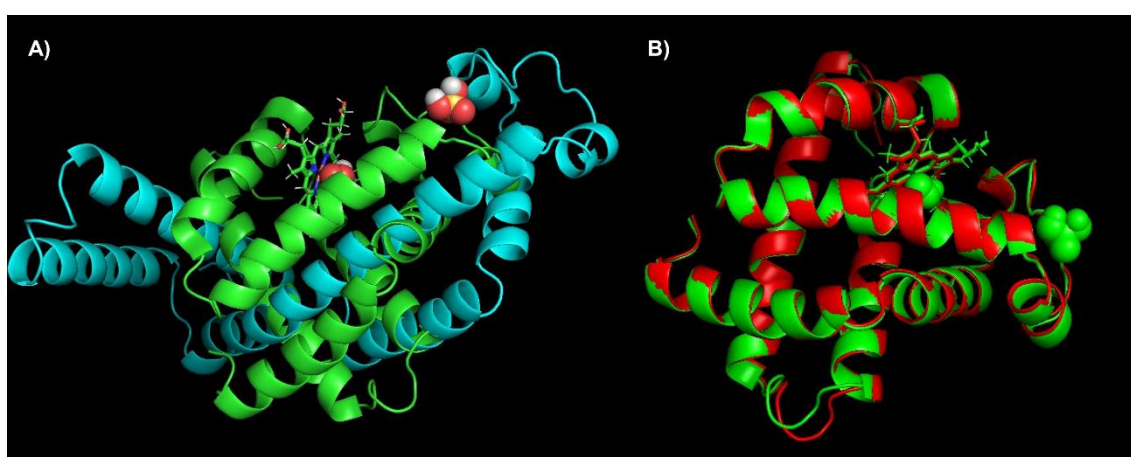


Figure 6 - Comparison between 1MBO crystal structure (green) and the SWISS-MODEL 3D structure predictions (cyan and red). **A)** RMSD = 19.654; **B)** RMSD = 0.396

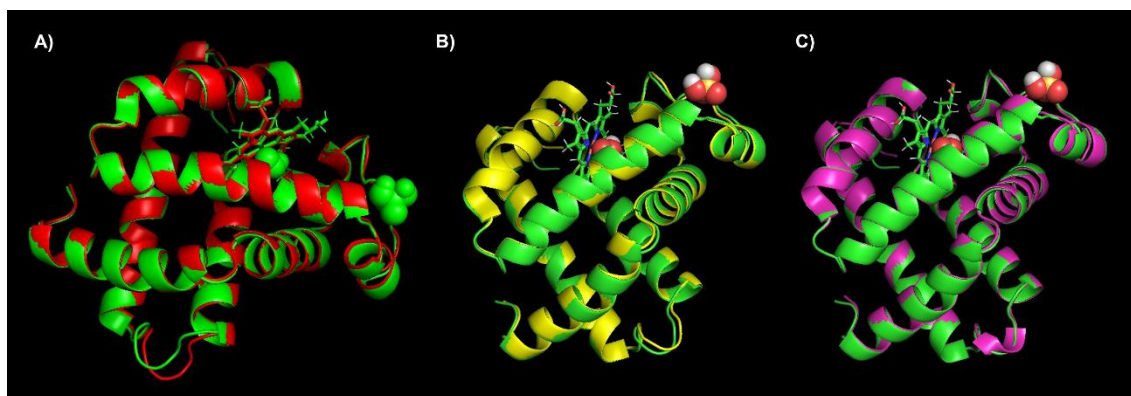


Figure 7 - Comparison between 1MBO crystal structure (green) and Predicted Structures. **A)** SWISS-MODEL (red) with RMSD = 0.396; **B)** ESM (yellow) with RMSD = 0.412; **C)** AlphaFold (pink) with RMSD = 0.381.

Molecular Docking

This docking process was performed as we mentioned in the abstract using the HADDOCK 2.4 tool. After doing the homology modelling, we used the best one (the one with the lowest RMSD) out of the three for this next process as well as our HEME ligand, which were prepared as input files. For this test, three different docking approaches were used: Active & Passive Residue list – this test consists in defining and specifying, previous to its execution, both the active and passive residues (passive, in case they exist) obtained from the previous interface analysis; Blind – in this test there are no constraint applied regarding the residues; Random Patches – in this last one, the binding residues are selected randomly (Table 2). Lastly for each of the approaches, we analysed the top 10 best complexes obtain in each test, where the analysis consisted in aligning all the 10 complexes to our crystal structure, calculate the RMSD values and evaluate the docking accuracy, after doing so, we chose only one model per 10 complexes that represented the best one from each of the tests, ending therefore, with three files only. With the best model out of those three being the one from the Active and Passive Residues test, which had the lowest RMSD (Figure 8).

Table 2 - Excel table with the three docking tests done, the 10 model complexes obtained for each of them, both the Haddock and RMSD scores, and the best model for each test.

Haddock								
Active and Passive Residues			Blind Docking			Random Patches Docking		
Cluster	HADDOCK	RMSD	Cluster	HADDOCK	RMSD	Cluster	HADDOCK	RMSD
model_1	-256.43	23.02	model1_1	-272.18	25.26	model2_1	-271.08	24.12
model_2	-221.8	23.11	model1_2	-243.94	25.44	model2_2	-242.54	24.33
model_3	-186.56	24.02	model1_3	-203.72	26.17	model2_3	-201.7	25.08
model_4	-182.02	22.26	model1_4	-199.09	25.04	model2_4	-198	23.71
model_5	-181.12	21.68	model1_5	-198.98	23.8	model2_5	-197.91	22.49
model_6	-164.23	25.31	model1_6	-181.34	27.59	model2_6	-180.34	26.41
model_7	-162.78	21.13	model1_7	-179.35	22.89	model2_7	-175.44	21.59
model_8	-158.9	21.91	model1_8	-175.79	23.6	model2_8	-174.23	22.34
model_9	-158.63	18.98	model1_9	-175.55	19.81	model2_9	-174.07	19.12
model_10	-153.79	20.62	model1_10	-170.51	22.76	model2_10	-169.16	21.41

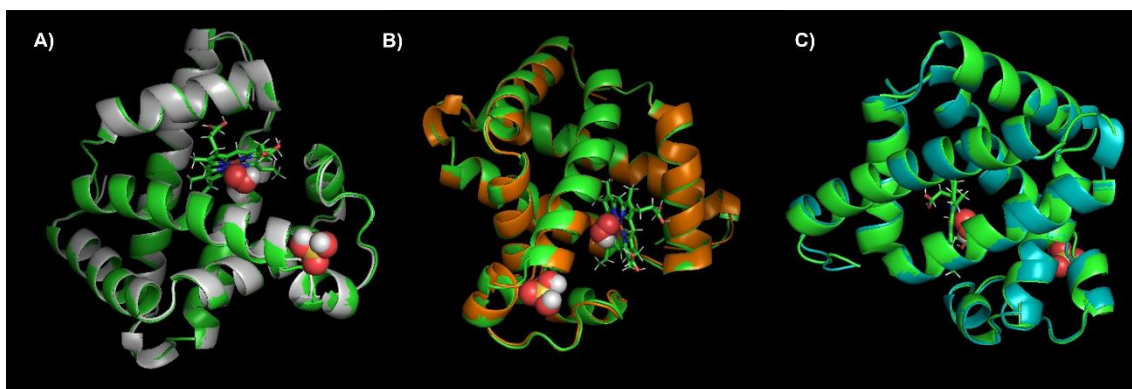


Fig. 8. Comparison between the best model for each of the docking tests and the crystal structure (green). **A)** Random Patches (white) with RMSD = 0.397; **B)** Active and Passive Residues (orange) with RMSD = 0.381; **C)** Blind (cyan) with RMSD = 0.416.

Hotspot Analysis

Before this last part, we compared the last three files obtained from the HADDOCK tests to get the ultimate best one to use in this following process, which ended up being the model obtained from the Active and Passive Residues docking test. Through the use of the Hotspot Wizard, we identified both Functional and Stability hotspots for the selected model. Those same hotspots, that were identified by this web server, were compared with the actual residues seen using PyMOL, to check which of them were present in the interface residues list obtained earlier. By looking and comparing the Stability Hotspots to our interface residues, we can see that, only five out of eighteen are correspondent to the ones that we identified in the interface of our HEME ligand: ASP 44 A, ARG 45 A, ILE 75 A, TYR 103 A, LEU 104 A (Figure 9). On the other hand, looking and comparing the Functional Hotspots to our interface residues, we can see that, out of the four identified hotspots, three of them are present in our list: LYS 42 A, ASP 44 A, THR 67 A (Figure 10).

	chain	position	residue
Chain A			
	A	118	Arg
	A	12	His
	A	45	Arg
	A	74	Ala
	A	151	Tyr
	A	132	Asn
	A	35	Ser
	A	117	Ser
	A	113	His
	A	28	Ile
	A	15	Ala
	A	110	Ala
	A	44	Asp
	A	104	Leu
	A	83	Glu
	A	75	Ile
	A	103	Tyr
	A	55	Met

R	P	C
LEU 32 A		ILE 75 A
PHE 33 A		PRO 88 A
THR 39 A		LEU 89 A
LEU 40 A		SER 92 A
LYS 42 A		HIS 93 A
PHE 43 A		LYS 96 A
ASP 44 A		HIS 97 A
ARG 45 A		ILE 99 A
PHE 46 A		PRO 100 A
HIS 64 A		TYR 103 A
THR 67 A		LEU 104 A
VAL 68 A		PHE 106 A
LEU 69 A		ILE 107 A
ALA 71 A		LEU 135 A
LEU 72 A		PHE 138 A

Figure 9 - Comparison between the identified Stability Hotspots and our Interface residues.

	chain	position	residue	R	P	C	R	P	C
				LEU 32 A			ILE 75 A		
				PHE 33 A			PRO 88 A		
				THR 39 A			LEU 89 A		
				LEU 40 A			SER 92 A		
				LYS 42 A			HIS 93 A		
				PHE 43 A			LYS 96 A		
				ASP 44 A			HIS 97 A		
				ARG 45 A			ILE 99 A		
				PHE 46 A			PRO 100 A		
				HIS 64 A			TYR 103 A		
				THR 67 A			LEU 104 A		
				VAL 68 A			PHE 106 A		
				LEU 69 A			ILE 107 A		
				ALA 71 A			LEU 135 A		
				LEU 72 A			PHE 138 A		

Figure 10 - Comparison between the identified Functional Hotspots and our Interface residue.

Conclusions

This project provided an overall comprehensive analysis of the 1MBO (myoglobin) protein using homology modelling and molecular docking to explore its structural and functional characteristics. Through careful preparation and analysis of our protein's structure, we were able to obtain key insights into the way our protein's chain A interacts with the HEME ligand. The homology modelling step revealed that, in comparison with the other tools, the AlphaFold2 predicted structure was the best one, having the highest accuracy when compared to the myoglobin's crystal structure, with an RMSD of 0.381. In terms of the molecular docking, this step highlighted that the Active and Passive Residue approach was the most effective, producing a model that closely aligned with our protein.

When it came to the Hotspot analysis, this helped further identify the functional and stability residues that are critical for our protein's interactions with the HEME ligand, reinforcing the importance of maintaining the binding stability of the interface residues. Although throughout this project we faced some technical problems such as the unavailability of certain tools like the CoCoMaps, issues with the use of other platforms such as GetArea, and Modeller, overall, we were still able to fulfil the purpose of this project and better understand the characteristics and the way the Myoglobin protein works and how it interacts with the HEME ligand.

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